

SHEAF EQUIVALENCES, POSTNIKOV TRUNCATIONS AND PREORDER REFLECTIONS IN THE LATTICE OF TWO-OUT-OF-THREE CLASSES

RYOTA SHIBUSAWA

ABSTRACT. For a category C let $\text{Same}(C)$ be the complete lattice of classes of morphisms that contain the identities and are closed under composition and two-out-of-three. Every reflector R contributes the element $W_R = \{f : Rf \text{ invertible}\}$, and the question studied here is how the lattice operations of $\text{Same}(C)$ interact with the reflectors that arise in sheaf theory, homotopy theory and order theory. Three results are proved. (1) In the simplicial objects of any ∞ -topos, the join of the realization equivalences, the $(n+1)$ -coskeletal equivalences and the levelwise equivalences of a left-exact localization L is exactly the class of maps inverted by the n -truncated L -sheaf of the realization; for simplicial sets this says that weak homotopy equivalences and coskeletal equivalences generate the Postnikov n -equivalences under two-out-of-three. (2) The sheaf equivalences W_J of the Grothendieck topologies J on a small category S form a meet-subsemilattice of $\text{Same}(\text{Presh}(S))$; joins are preserved for finite acyclic categories and for finite categories whose two topologies have an idempotent intersection of ideals, but not in general: on the poset $\mathbb{N}$ the join of two sheaf classes is a proper class inverted by a pro-limit. For finite categories the tower of bimodules representing the alternating plus-construction tower is pro-isomorphic to the bimodule representing the join sheafification, so that the plus tower has the join sheafification as its colimit and this obstruction disappears; yet the tower need not stabilize: an explicit monoid of order 12 has a non-stabilizing tower, and a conjectured three-fold multiplicativity of tensor products of ideals is refuted by the contracted monoid of a diamond poset. (3) The morphisms $f : X \rightarrow Y$ admitting a morphism $Y \rightarrow X$ form the largest element of $\text{Same}(C)$ inverted by the preorder reflection; localizing at it is the reflection into EI-categories, which for categories with binary products or coproducts is the preorder reflection itself, so that the EI-reflection of the category of graphs is the homomorphism preorder, whose partial-order reflection is the homomorphism order.

2020 *Mathematics Subject Classification.* 18A32 (primary), 18F10, 18F20, 18N60, 55P60, 06B23, 20M12, 05C60 (secondary).

Key words and phrases. two-out-of-three, localization, Grothendieck topology, subtopos, sheafification, Postnikov truncation, coskeleton, ∞ -topos, lex modality, EI-category, preorder reflection, homomorphism order of graphs, finite monoid, tensor product of bisets.

Corresponding author. E-mail: r-shibusawa@daiichi-koudai.ac.jp.

1. INTRODUCTION

A *sameness* on a category C is a class W of morphisms containing the identities and closed under composition and two-out-of-three: if two of f , g , gf lie in W , so does the third. Weak equivalences of model categories, the equivalences inverted by a reflector, the local isomorphisms of a Grothendieck topology, and the morphisms with a morphism back are examples. Samenesses form a complete lattice $\text{Same}(C)$ under inclusion [14], and every functor $F: C \rightarrow D$ contributes the element $F^*(\text{Iso}) = \{f : Ff \text{ invertible}\}$, the largest sameness that F inverts. This paper studies the joins of such elements for the reflectors of sheaf theory, homotopy theory and order theory. The join of two samenesses is the smallest sameness containing both; it is in general much smaller than the class of maps inverted by the composite reflector, and the results below identify when the two coincide. Three theorems are proved.

Theorem 1.1 (Postnikov truncation as a join). *Let $\mathcal{E}$ be an ∞ -topos, $\mathcal{P} = \text{Fun}(\Delta^{\text{op}}, \mathcal{E})$ its simplicial objects, $|-|$ the geometric realization, cosk_{n+1} the coskeleton, L a left-exact localization of $\mathcal{E}$ applied levelwise. In $\text{Same}(\mathcal{P})$, for every $n \geq -1$,*

$$W_{|-|} \vee W_{\text{cosk}_{n+1}} \vee W_L = W_{L\tau_{\leq n}|-|}.$$

For simplicial sets: $W_{\text{test}} \vee W_{\text{cosk}_{n+1}} = T_n$, where for $n \geq 0$ T_n is the class of maps inducing isomorphisms on π_i for $i \leq n$ at all base points, and T_{-1} is the class of maps $X \rightarrow Y$ with $X = \emptyset$ iff $Y = \emptyset$.

Theorem 1.2 (topologies in the sameness lattice). *For a small category S the assignment $J \mapsto W_J = \{f : a_J f \text{ iso}\}$ is an order-embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$ preserving all meets. It preserves binary joins for finite acyclic categories and for finite categories whose topologies have an idempotent intersection of ideals, but not for the poset $\mathbb{N}$. For every finite category the tower of bimodules representing the alternating plus-construction tower is pro-isomorphic to the bimodule representing the join sheafification, so the plus tower has colimit the join sheafification; the tower need not stabilize on the nose.*

Theorem 1.3 (the cycle class). *For every category C the morphisms with a morphism back form the largest sameness W_{cyc} inverted by the preorder reflection of C , and localizing at W_{cyc} is the reflection into EI-categories. If C has binary products or binary coproducts, $C[W_{\text{cyc}}^{-1}]$ is the reachability preorder of C ; for the category of graphs it is the homomorphism preorder, identifying finite graphs with isomorphic cores.*

What is known. The lattice $\text{Same}(C)$, its Galois functoriality and the comparison intervals $[\text{Iso}, F^*(\text{Iso})]$ are from [14]. The embedding of topologies, the finite-sieve dictionary between topologies and idempotent ideals (for finite S), and the simplicial-set case of Theorem 1.1 are from [15]; meets, the two positive join theorems and the $\mathbb{N}$ counterexample from [16]; the cycle

class and its localization from [17]; and the ∞ -topos form of Theorem 1.1 from [18]. The facts about localizations of ∞ -topoi are in [11, 1]; sheaf theory follows [13, 6]; simplicial homotopy theory [4]; EI-categories [7]; the homomorphism order of graphs [5].

Related work. The lattice of Grothendieck topologies, or of subtoposes, is classical [13, 6]; its description by idempotent ideals for essential localizations goes back to Kelly and Lawvere [9], and rigid topologies on finite and Cauchy-complete categories, including the simplex category, are characterized in [12]. The levels of the toposes of simplicial and cubical sets, i.e. the coskeletal essential subtoposes, are determined in [10]. For ∞ -topoi, Anel, Biedermann, Finster and Joyal describe left-exact localizations by congruences and Grothendieck topologies [1, 2]. All of these concern the lattice of *saturated* classes, i.e. of localizations themselves. The focus here is different: the bare two-out-of-three join of two such classes, before any saturation, and the question of when it is already the class of a reflector. Theorems 1.1 and 1.2 answer it positively in the homotopical setting and for two families of finite categories, and Section 6 shows what goes wrong in general.

Provenance. Proposition 2.2 is from [14]; Lemma 2.4, Theorem 2.5, Theorem 3.1 (first part), Lemma 4.1 and Theorem 4.2 from [15]; Theorem 3.1 (meets), Lemma 3.2, Theorems 3.3 and 3.5, Lemma 3.4 and Theorem 6.1 from [16]; Proposition 4.3 from [18]; and Section 5 from [17]. Proofs of all these results are included, so that the paper is self-contained; the remaining statements are new.

What is new. This paper supersedes and consolidates the preprints [14, 15, 16, 17, 18]. Besides the integration, the paper contains three results not in the cited preprints: the three-fold multiplicativity conjecture of [16, Conj. 6.12] is false (Proposition 6.2); for every finite category the representing tower of the alternating plus construction is pro-isomorphic to the representing object of the join sheafification, so that the plus tower has the join sheafification as its colimit (Theorem 6.4, Corollary 6.5), which shows that the pro-limit invariant of the $\mathbb{N}$ counterexample cannot occur for finite categories; and there are finite monoids whose alternating tower does not stabilize (Proposition 6.6), so that the finite join question is sharper than the stabilization criterion and remains open. The proofs are self-contained.

Outline. Section 2 sets up the lattice and the join theorem. Sections 3, 4 and 5 prove Theorems 1.2, 1.1 and 1.3. Section 6 collects the limits: the $\mathbb{N}$ example, the refuted conjecture, essential constancy and the oscillating tower, and the open problems.

2. THE LATTICE OF SAMENESSES

Definition 2.1. A *sameness* of a category C is a class $W \subseteq \text{Mor}(C)$ containing the identities, closed under composition, and satisfying two-out-of-three [14, Def. 2.1]. $\text{Same}(C)$ is the set of samenesses ordered by inclusion. For an ∞ -category the same definition is applied to the morphisms of its homotopy category. The isomorphisms form a sameness $\text{Iso}(C)$, and every class of the form $F^*(\text{Iso})$ below contains it; but a sameness need not contain non-identity isomorphisms (Section 5).

Proposition 2.2. $\text{Same}(C)$ is a complete lattice: meets are intersections, the top is $\text{Mor}(C)$, the bottom is $\{\text{id}\}$, and the join of a family is the sameness generated by its union. For a functor $F: C \rightarrow D$ and $W \in \text{Same}(D)$ the class $F^*(W) = \{f : Ff \in W\}$ is a sameness, and $F^*(\text{Iso})$ is the largest sameness inverted by F ; F^* preserves meets and has a left adjoint $F_!$, so that Same is a functor from $\mathbf{Cat}$ to complete lattices with Galois connections.

Proof. The three axioms are preserved by intersections, and $\text{Mor}(C)$ satisfies them, so $\text{Same}(C)$ is a complete lattice with joins the meets of upper bounds. For $F^*(W)$: F preserves identities and composition, and if two of f, g, gf lie in $F^*(W)$ then two of $Ff, Fg, F(gf)$ lie in W , hence the third. A functor inverts a sameness V iff $V \subseteq F^*(\text{Iso})$. Meet preservation is $F^*(\bigcap W_i) = \bigcap F^*(W_i)$, and a meet-preserving map of complete lattices has a left adjoint. $\square$

Definition 2.3. For $W \in \text{Same}(C)$ an object Z is W -local if $\text{Hom}(f, Z)$ is bijective for all $f \in W$; $\text{Loc}(W)$ is the class of local objects and $\text{Sat}(W) := \{f : \text{Hom}(f, Z) \text{ bijective for all } Z \in \text{Loc}(W)\} \supseteq W$ the *saturation*. W is *saturated* if $W = \text{Sat}(W)$.

Lemma 2.4. $\text{Loc}(U \vee V) = \text{Loc}(U) \cap \text{Loc}(V)$. If R is a reflector onto a full replete subcategory $\mathcal{R}$ then $W_R := R^*(\text{Iso})$ is saturated with $\text{Loc}(W_R) = \mathcal{R}$; consequently, if $\text{Loc}(W)$ is reflective with reflector R , then $\text{Sat}(W) = W_R$.

Proof. $\{f : \text{Hom}(f, Z) \text{ bijective}\}$ is a sameness, so Z is local for $U \vee V$ iff it is local for U and for V . For a reflector with unit η : Rf is invertible iff $\text{Hom}(Rf, Z)$ is bijective for all $Z \in \mathcal{R}$ iff $\text{Hom}(f, Z)$ is bijective for all $Z \in \mathcal{R}$, so $W_R = \{f : \text{Hom}(f, Z) \text{ bij. } \forall Z \in \mathcal{R}\}$ and $\mathcal{R} \subseteq \text{Loc}(W_R)$; if Z is W_R -local then $\eta_Z \in W_R$ gives a retraction r of η_Z , and $\text{Hom}(\eta_Z, RZ)$ bijective forces $\eta_Z r = \text{id}$, so $Z \in \mathcal{R}$. The last assertion is the description of W_R by orthogonality to $\mathcal{R} = \text{Loc}(W)$. $\square$

The saturation of Definition 2.3 is orthogonality to the local objects. It is in general larger than the class of maps inverted by the localization functor $C \rightarrow C[W^{-1}]$: Theorem 6.1 below gives a sameness whose local objects are the terminal objects only, whose saturation in this sense is therefore Mor , but which is inverted by a limit functor that does not invert every map.

Theorem 2.5 (join theorem). *Let $U, V, W \in \text{Same}(C)$ with $W \subseteq U \vee V$, and let $P: C \rightarrow C$ be a functor with a natural transformation $\pi: \text{id} \Rightarrow P$ such that (a) $\pi_X \in U \vee V$ for every X and (b) $P(U) \cup P(V) \subseteq W$. Then $U \vee V = P^*(W)$.*

Proof. $P^*(W)$ is a sameness containing U and V by (b), hence $U \vee V$. If $Pf \in W$, the naturality square $\pi_Y f = Pf \pi_X$ has three of its four sides in $U \vee V$, so $f \in U \vee V$ by two-out-of-three. $\square$

Corollary 2.6 (two reflectors). *For reflectors $R_1, R_2: W_{R_1} \vee W_{R_2} = W_{R_2 R_1}$ iff $R_2 R_1$ inverts W_{R_2} ; in particular whenever R_1 preserves R_2 -equivalences.*

Proof. Apply the theorem with $P = R_2 R_1$, $W = \text{Iso}$ and π_X the composite of the two units, a W_{R_1} -map followed by a W_{R_2} -map. Conversely the equality gives $W_{R_2} \subseteq W_{R_2 R_1}$. $\square$

3. SHEAF EQUIVALENCES

Let S be a small category, $\text{Top}(S)$ the complete lattice of Grothendieck topologies (ordered by inclusion of covering sieves), a_J the sheafification for J , a left-exact reflector onto $\text{Sh}_J(S)$, and $W_J := a_J^*(\text{Iso})$, the class of J -local isomorphisms [13, III.7].

Theorem 3.1 ([15, Thm. 3.1], [16, Thm. 3.1]). *$J \mapsto W_J$ is an order-embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$ which preserves arbitrary meets: $W_{\bigwedge J_i} = \bigcap W_{J_i}$. Its image consists of the saturated samenesses whose local objects form a subtopos. For all J, J' , $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'})$, so $W_J \vee W_{J'} \subseteq W_{J \vee J'}$ with equality iff the join is saturated.*

Proof. W_J is a sameness by Proposition 2.2, saturated with $\text{Loc}(W_J) = \text{Sh}_J$ by Lemma 2.4; the map is injective and monotone because $J \leq J'$ iff $\text{Sh}_{J'} \subseteq \text{Sh}_J$, and a saturated sameness is determined by its local objects. For meets: a map f is a J -local isomorphism iff for every object c and every $x \in X(c)$, $y \in Y(c)$ the sieves $\{u : X(u)x = X(u)x'\}$ (for x' with $fx = fx'$) and $\{u : Y(u)y \in \text{im} f\}$ are J -covering; these sieves do not depend on J , and a sieve is covering for $\bigwedge J_i$ iff it is covering for every J_i ; hence $W_{\bigwedge J_i} = \bigcap W_{J_i}$. For the image: a saturated sameness whose local objects are a subtopos Sh_J equals W_J by Lemma 2.4. Finally $\text{Loc}(W_J \vee W_{J'}) = \text{Sh}_J \cap \text{Sh}_{J'} = \text{Sh}_{J \vee J'}$, so the saturation is $W_{J \vee J'}$. $\square$

3.1. The finite dictionary. Call a two-sided ideal D of S (a class of morphisms closed under composition with arbitrary morphisms on both sides) *idempotent* if every member is a composite of two members. When S has finitely many sieves on each object, $\text{Top}(S) \cong \text{Idem}(S)^{\text{op}}$ via $J \mapsto D_J = (\bigcap J(c))_c$ and $D \mapsto J_D(c) = \{R \supseteq D(c)\}$ [15, Thm. 5.1]. For a finite monoid M the topologies are thus the idempotent two-sided ideals under reverse inclusion; for a finite acyclic category (no cycles, no non-identity endomorphisms) they are the subsets Y of objects, D_Y the morphisms factoring through Y , $\text{Sh}_{J_Y} \simeq \text{Presh}(Y)$ and $W_Y = \{f : f_y \text{ bijective for } y \in Y\}$

[16, §5]. The join $J_D \vee J_E$ corresponds to the largest idempotent ideal I' contained in $D \cap E$; for a finite monoid $I' = (DE)^m$ for m large.

3.2. The alternating plus-construction tower. For topologies J, J' write $\Phi_k X$ for the k -fold alternating plus construction $X^{+J+J'+J\cdots}$, with units $u_k: X \rightarrow \Phi_k X$, each a composite of maps of $W_J \cup W_{J'}$. For a finite monoid M with idempotent ideals D, E (viewed as M - M -bisets), $X^{+D} = \text{Hom}_M(D, X)$ and $\Phi_k X = \text{Hom}_M(T_k, X)$ with $T_k = D_k \otimes_M \cdots \otimes_M D_1$ the alternating tensor tower ($D_1 = D, D_2 = E, \dots$), the units being induced by the multiplications $\mu_k: T_k \rightarrow T_{k-1}$; and $a_D X = \text{Hom}_M(D \otimes D, X)$.

Lemma 3.2 (reduction). *If for every X there is $k = k(X)$ such that $\Phi_k X$ is a $(J \vee J')$ -sheaf, then $W_J \vee W_{J'} = W_{J \vee J'}$. For a finite monoid, the tower is uniformly stable, i.e. there is k_0 with $\Phi_k X \rightarrow \Phi_{k+1} X$ bijective for all X and all $k \geq k_0$, iff μ_k is bijective for all $k \geq k_0$; either condition implies join preservation.*

Proof. For $f: X \rightarrow X'$ in $W_{J \vee J'}$ take k working for both; $\Phi_k f$ is a map of $(J \vee J')$ -sheaves inverted by $a_{J \vee J'}$, hence an isomorphism, and $u_{X'} f = \Phi_k f u_X$ gives $f \in W_J \vee W_{J'}$. By Yoneda, $\text{Hom}(\mu_{k+1}, X)$ is bijective for all X iff μ_{k+1} is; and if the units are eventually bijective then $\Phi_{k_0} X$ is a J -sheaf and a J' -sheaf, hence a $(J \vee J')$ -sheaf, for every X . $\square$

Theorem 3.3 (finite acyclic categories). *For a finite acyclic category S and all $Y, Y' \subseteq \text{Ob} S$, $W_Y \vee W_{Y'} = W_{Y \cap Y'}$; thus $\text{Top}(S) \cong (2^{\text{Ob} S})^{\text{op}}$ is a sublattice of $\text{Same}(\text{Presh}(S))$.*

Proof. Let $f: X \rightarrow X'$ be bijective on $Y \cap Y'$. Choose a linear extension $p_1, \dots, p_m$ of the reachability preorder and put $S_i = \{p_1, \dots, p_i\}$. Define the mixed presheaf X_i by $X_i(q) = X'(q)$ for $q \in S_i$ and $X(q)$ otherwise, with restriction along $u: q \rightarrow q'$ equal to $X'(u)$, $X(u)$, or $f_q \circ X(u)$ according as both, neither, or only q lies in S_i (the remaining case cannot occur since S_i is closed under predecessors); functoriality is the naturality of f . The maps $\varphi_i: X_{i-1} \rightarrow X_i$ (identity away from p_i , f_{p_i} there) are natural, $f = \varphi_m \cdots \varphi_1$, and φ_i is bijective away from p_i : it lies in W_Y if $p_i \notin Y$, in $W_{Y'}$ if $p_i \notin Y'$, and is an isomorphism if $p_i \in Y \cap Y'$. Hence $f \in W_Y \vee W_{Y'}$; the reverse inclusion is Theorem 3.1. $\square$

For finite monoids the argument is replaced by the tensor tower. Two facts about a proper idempotent ideal D of a finite monoid are used: every $d \in D$ can be written $d = pgq$ with $p \in D, g \in D$ idempotent, $pg = p$ (write d as a long product of elements of D ; two prefixes coincide, $p_i = p_i w$, and $g = w^\omega$), and symmetrically $d = p'g'q'$ with $q' \in D, g'q' = q'$.

Lemma 3.4 (merging). *Let X, Y, Z be two-sided ideals of a finite monoid with Y idempotent, and consider $X \otimes Y \otimes Z \otimes \cdots$ (further slots unaffected).*

- (P) *If $c \in Y \cdot Z$, the class $[x, y, c, \dots]$ depends only on $(xy, c, \dots)$. (P')*
- Mirror image: if $x \in X \cdot Y$, the class $[x, y, c, \dots]$ depends only on $(x, yc, \dots)$.*

- (A) (*Absorption*) If the ideal of a slot is idempotent, an element w of that slot may be replaced by w' with $w = w'w''$, the factor w'' being multiplied into the next slot to the right; so an interior slot may be assumed to lie in the product of its ideal with the ideal to its right.
- (M) If the conclusion of (P) holds for all tuples, then the multiplication $X \otimes Y \otimes Z \otimes \cdots \rightarrow XY \otimes Z \otimes \cdots$ is bijective; likewise for (P') and $X \otimes YZ \otimes \cdots$.

Proof. (P) Write $c = y_1 z_1$ with $y_1 \in Y$, $z_1 \in Z$, and $y_1 = y'gy''$ with g idempotent in Y , $y'g = y'$ (prefix factorization); moving $y'g$ to the left and then yy' to the left, $[x, y, c] = [x, yy'g, y''z_1] = [xyy', g, y''z_1]$, each move legitimate because the remaining entries lie in their slot ideals; the result depends on xy and on data determined by c . (P') is the mirror image, with the suffix factorization. (A) is the defining relation of the tensor product. (M) The map is surjective; $[w, c, \dots] \mapsto [x, y, c, \dots]$ for any factorization $w = xy$ is well defined by hypothesis and respects the relations of the target (for $[wm, c] \sim [w, mc]$ use the factorization $wm = x(y m)$), so it is a section, and μ is injective. $\square$

Theorem 3.5 (idempotent intersections). *Let M be a finite monoid and D, E idempotent two-sided ideals with $I := D \cap E$ idempotent. Then $T_k \cong I \otimes_M I$ for $k \geq 4$ by the multiplication maps, $\Phi_4 X \cong a_I X$ for every M -set X , and $W_D \vee W_E = W_{D \cap E}$. The same holds for finite categories, with profunctor tensors.*

Proof. Every product of two or more alternating factors D, E lies between $I = I^2$ and $D \cap E = I$, hence equals I . Write the tuples of T_k as $[x_1, \dots, x_k]$ with slot ideals $D_k, \dots, D_1$ from left to right.

Step 1: merging slots 1, 2. Let t, t' be tuples differing only in slots 1, 2 with $x_1 x_2 = x'_1 x'_2$. Absorb slot 4 into slot 3 in both, with the same factorization $x_4 = pq$ (Lemma 3.4(A); here $k \geq 4$ is used): slot 3 becomes $x_3 p \in D_{k-2} D_{k-3} = I \subseteq D_{k-1} D_{k-2}$. Lemma 3.4(P) for the slots 1, 2, 3 shows that the classes of t and t' agree, and (M) gives a bijection $T_k \rightarrow I \otimes D_{k-2} \otimes \cdots \otimes D_1$.

Step 2: merging slots 2, 3 repeatedly. In $I \otimes Y \otimes Z \otimes \cdots$ with Y idempotent every element of the first slot lies in $I = I \cdot I \subseteq I \cdot Y$, so (P') applies to every tuple and (M) gives a bijection onto $I \otimes YZ \otimes \cdots$; since $YZ = I$ and I is idempotent this repeats until two slots remain. All maps are multiplications, so the composite $T_k \rightarrow I \otimes I$ is the canonical one and is an isomorphism of bisets.

Hence $\Phi_k X = \text{Hom}_M(T_k, X) \cong \text{Hom}_M(I \otimes I, X) = a_I X$, an I -sheaf, and Lemma 3.2 gives $W_D \vee W_E = W_I$.

Finite categories. For a finite category S , ideals are sub-profunctors $D \subseteq \text{Hom}_S$, the tensor is the composite profunctor (classes of composable chains modulo moving a morphism across a tensor sign), $X^{+D} = \text{Hom}(D, X)$ with $\text{Hom}(P, X)(c) = \text{Hom}_{\text{Presh}(S)}(P(-, c), X)$, and $a_D X = \text{Hom}(D \otimes_S D, X)$. The factorization fact used in Lemma 3.4 holds for an idempotent ideal Y

of S : writing $y \in Y(a, c)$ as a composite of more members of Y than S has morphisms, two partial composites with domain a coincide, $p_i = p_j = m \circ p_i$ with $m \in Y$ an endomorphism, and a power $g = m^r$ is an idempotent of Y with $g \circ p_i = p_i$; this gives $y = y' \circ g \circ y''$ with $g \circ y'' = y''$, and dually. The proofs of Lemma 3.4 and of Steps 1–2 use only the tensor relations, closure of ideals under composition and this factorization, so they apply verbatim. $\square$

The hypothesis fails already for the chain $a < b$ ($D_{\{a\}} \cap D_{\{b\}}$ is not idempotent), which is covered by Theorem 3.3 instead. What happens beyond both theorems is the subject of Section 6.

4. POSTNIKOV TRUNCATIONS

4.1. Coskeletal equivalences. For the simplex category Δ (and the cube category with faces and degeneracies) the two-sided ideals are $\emptyset$, the ideals D_k of morphisms factoring through an object of dimension $\leq k$, and Mor ; all are idempotent, so the topologies form a chain and the J_{D_k} -sheaves are the k -coskeletal presheaves [15, Thm. 6.1]. The sheaf samenesses are the coskeletal equivalences $W_{D_k} = \{f : \text{cosk}_k f \text{ iso}\} = \{f : f_m \text{ bijective for } m \leq k\}$, a descending chain with $\bigcap_k W_{D_k} = \text{Iso}$.

Lemma 4.1. *For a Kan complex K and $n \geq -1$, $\text{cosk}_{n+1}K$ is a Kan complex, the unit $K \rightarrow \text{cosk}_{n+1}K$ induces isomorphisms on π_i for $i \leq n$ at all base points, and $\pi_i(\text{cosk}_{n+1}K) = 0$ for $i \geq n+1$ (for $n = -1$: empty or contractible).*

Proof. An m -simplex of $\text{cosk}_{n+1}K$ is a map $\text{sk}_{n+1}\Delta^m \rightarrow K$. A horn $\Lambda_m^k \rightarrow \text{cosk}_{n+1}K$ is a map $\text{sk}_{n+1}\Lambda_m^k \rightarrow K$; for $m \geq n+3$ this skeleton is $\text{sk}_{n+1}\Delta^m$ and the horn is already a simplex, for $m \leq n+2$ it is a horn in K whose filler restricts to the required simplex. Homotopy groups of a Kan complex are computed from simplices of dimension $i, i+1$ with boundary at the base point, where the unit is bijective for $i \leq n$; for $i \geq n+1$ every map $\partial\Delta^{i+1} \rightarrow \text{cosk}_{n+1}K$ is a map $\text{sk}_{n+1}\Delta^{i+1} \rightarrow K$, i.e. extends to Δ^{i+1} , so $\pi_i = 0$. $\square$

Theorem 4.2. *In $\text{Same}(\mathbf{sSet})$, with W_{test} the weak equivalences, $T_n = \{f : \pi_i f \text{ iso for } i \leq n \text{ at all base points}\}$ for $n \geq 0$ and $T_{-1} = \{f : X \rightarrow Y : X = \emptyset \Leftrightarrow Y = \emptyset\}$: $W_{\text{test}} \vee W_{D_{n+1}} = T_n$ for all $n \geq -1$, and $W_{\text{test}} = \bigcap_n (W_{\text{test}} \vee W_{D_n})$.*

Proof. Apply Theorem 2.5 with $P = \text{cosk}_{n+1}\text{Ex}^\infty$ and $W = W_{\text{test}}$: π_X is the Kan replacement followed by a coskeleton unit, P preserves weak equivalences (Lemma 4.1 and Whitehead's theorem for Kan complexes), and for $f \in W_{D_{n+1}}$, $\text{Ex}^\infty f$ is again bijective in dimensions $\leq n+1$ (as Ex^∞ is computed dimensionwise from the same low-dimensional data), so Pf is a weak equivalence by Lemma 4.1. Thus the join is $P^*(W_{\text{test}}) = T_n$ by Lemma 4.1. The intersection statement is Whitehead's theorem. $\square$

4.2. Simplicial objects of an ∞ -topos. Let $\mathcal{E}$ be an ∞ -topos, $\mathcal{P} = \text{Fun}(\Delta^{\text{op}}, \mathcal{E})$, $|-| = \text{colim}$, c the constant functor (fully faithful, since Δ^{op} has an initial object), cosk_{n+1} the coskeleton (a right Kan extension computed by finite limits, hence commuting with every functor preserving finite limits), and $\tau_{\leq n}$ the truncation. $W_{|-|}$ and $W_{\text{cosk}_{n+1}}$ are the saturated samenesses of the reflectors $c|-|$ and cosk_{n+1} . A constant object cK is $(n+1)$ -coskeletal iff K is n -truncated, since the unit at level $n+2$ is $K \rightarrow K^{S^{n+1}}$ [16, Lemma 8.2]; but the levels $K^{|\text{sk}_{n+1}\Delta^m|}$ of $\text{cosk}_{n+1}cK$ are not truncated. The key fact is that its realization is.

Proposition 4.3. *For $K \in \mathcal{E}$ and $n \geq -1$ the realization of the coskeleton unit $cK \rightarrow \text{cosk}_{n+1}cK$ is the truncation $K \rightarrow \tau_{\leq n}K$; so $|\text{cosk}_{n+1}cK| \simeq \tau_{\leq n}K$ naturally in K .*

Proof. For $\mathcal{E} = \mathcal{S}$ choose a Kan complex K' with $|K'| \simeq K$ and consider the bisimplicial sets $S_{m,k} = \text{Hom}(\Delta^m \times \Delta^k, K')$, $T_{m,k} = \text{Hom}(\text{sk}_{n+1}\Delta^m \times \Delta^k, K')$ and $T'_{m,k} = \text{Hom}(\Delta^m \times \Delta^k, \text{cosk}_{n+1}K')$ with the restriction maps $S \rightarrow T \rightarrow T'$. For fixed m the columns $S_{m,\bullet} = K'^{\Delta^m}$ and $T_{m,\bullet} = K'^{\text{sk}_{n+1}\Delta^m}$ are simplicial mapping spaces into a Kan complex, of homotopy types K and $\text{Map}(|\text{sk}_{n+1}\Delta^m|, K)$, so $S \rightarrow T$ models the coskeleton unit $cK \rightarrow \text{cosk}_{n+1}cK$, and the realization of a simplicial space obtained by realizing a bisimplicial set columnwise is the diagonal of the bisimplicial set [4, IV.1.9 and Ex. IV.1.7]; thus the realized unit is $\text{diag}S \rightarrow \text{diag}T$. For fixed k the rows are $K'^{\Delta^k} \rightarrow \text{cosk}_{n+1}(K'^{\Delta^k}) \rightarrow (\text{cosk}_{n+1}K')^{\Delta^k}$; by Lemma 4.1 the first map is an n -equivalence to an n -truncated Kan complex, and so is the composite (the exponential of the unit of K'), whence the second map is a weak equivalence. A map of bisimplicial sets that is a weak equivalence in each row induces a weak equivalence of diagonals [4, IV.1.7]; applied to $T \rightarrow T'$ and to the rowwise equivalences $K'^{\Delta^k} \rightarrow K'$ and $(\text{cosk}_{n+1}K')^{\Delta^k} \rightarrow \text{cosk}_{n+1}K'$ (evaluation at the vertex 0 of Δ^k , which has a section and is a weak equivalence of Kan complexes), this identifies $\text{diag}S \simeq K'$ and $\text{diag}T \simeq \text{diag}T' \simeq \text{cosk}_{n+1}K'$ compatibly with the units, so that $\text{diag}S \rightarrow \text{diag}T$ is the coskeleton unit of K' , i.e. the Postnikov section by Lemma 4.1. For a presheaf ∞ -topos everything is computed objectwise, and for $\mathcal{E} = L\text{Presh}(C)$ the coskeleton is computed in $\text{Presh}(C)$ while $|-|_{\mathcal{E}} = L|-|$ and $L\tau_{\leq n} = \tau_{\leq n}L$ [11, 5.5.6.28], so the identification is transported by L . $\square$

Proof of Theorem 1.1. First $W_{|-|} \vee W_{\text{cosk}_{n+1}} = W_{\tau_{\leq n}|-|}$: apply Theorem 2.5 with $P = c\tau_{\leq n}|-|$ and $\pi_X: X \rightarrow c|X| \rightarrow c\tau_{\leq n}|X|$. Condition (b): P inverts $W_{|-|}$, and inverts $W_{\text{cosk}_{n+1}}$ because the local objects of the join are the constant objects on n -truncated objects, a reflective subcategory with reflector P , so P inverts the whole join. Condition (a): the first map is the unit of $|-| \dashv c$, in $W_{|-|}$; the second is $c(K \rightarrow \tau_{\leq n}K)$ for $K = |X|$, and by Proposition 4.3 it is, up to the equivalence $c|\text{cosk}_{n+1}cK| \simeq c\tau_{\leq n}K$, the composite of the coskeleton unit $cK \rightarrow \text{cosk}_{n+1}cK$ (in $W_{\text{cosk}_{n+1}}$) with the unit of

$|-| \dashv c$ at $\text{cosk}_{n+1}cK$ (in $W_{|-|}$). Then apply Corollary 2.6 to the reflectors $c\tau_{\leq n}|-|$ and levelwise L : their composite inverts W_L because L commutes with truncation and with colimits. For simplicial sets, Theorem 4.2. $\square$

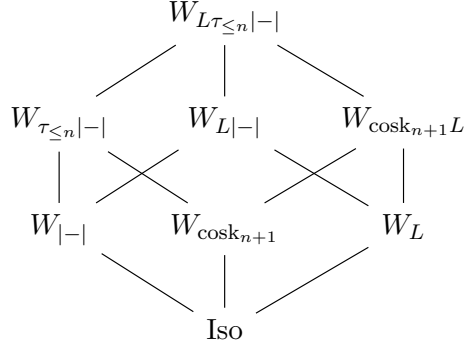


FIGURE 1. Theorem 1.1: the sublattice of $\text{Same}(\text{Fun}(\Delta^{\text{op}}, \mathcal{E}))$ generated by the three reflector classes; every join is again a reflector class (lines are joins, read upwards).

5. THE CYCLE CLASS AND PREORDER REFLECTION

Write $X \preceq Y$ if $\text{Hom}(X, Y) \neq \emptyset$ (the reachability preorder of C) and let $W_{\text{cyc}}(C) = \{f: X \rightarrow Y : Y \preceq X\}$, the morphisms with a morphism back. A category is an *EI-category* if all its endomorphisms are invertible [7].

Theorem 5.1. *$W_{\text{cyc}}(C)$ is a sameness, equal to $P^*(\text{Iso})$ for the preorder reflection $P: C \rightarrow (\text{Ob}C, \preceq)$; it is $\{\text{id}\}$ iff C is acyclic, Iso iff C is an EI-category, and Mor iff $\preceq$ is symmetric. The localization $C[W_{\text{cyc}}^{-1}]$ is an EI-category with the same reachability preorder, and $C \rightarrow C[W_{\text{cyc}}^{-1}]$ is universal among functors to EI-categories.*

Proof. Pf is an isomorphism in the preorder iff $X \preceq Y \preceq X$ iff f has a back, so $W_{\text{cyc}} = P^*(\text{Iso})$ is a sameness. The three extremes are immediate. For the localization: a zigzag $X \rightarrow \cdots \rightarrow Y$ in $C[W_{\text{cyc}}^{-1}]$ exists iff $X \preceq Y$ (backward arrows are in W_{cyc} , hence reversible in $\preceq$), so the preorder is preserved; an endomorphism of X in the localization is a zigzag whose forward arrows all lie in strongly connected classes reachable from and reaching X , so each forward arrow has a back and is inverted; hence the zigzag is invertible and the localization is EI. Conversely a functor to an EI-category D sends $f: X \rightarrow Y$ with back g to $F(f)$ with $F(g)F(f)$ and $F(f)F(g)$ invertible endomorphisms, so $F(f)$ is invertible; so F inverts W_{cyc} and factors uniquely through the localization. $\square$

Theorem 5.2. *If C has binary coproducts or binary products, $C[W_{\text{cyc}}^{-1}]$ is the reachability preorder $(\text{Ob}C, \preceq)$: the EI-reflection is the preorder reflection.*

Proof. For parallel $g, g': B \rightarrow C'$ let $j_1, j_2: B \rightarrow B \sqcup B$ be the injections and ∇ the fold map, with $\nabla j_1 = \nabla j_2 = \text{id}$. Then ∇ has the back j_1 , so it is inverted in $C[W_{\text{cyc}}^{-1}]$, where $j_1 = \nabla^{-1} = j_2$ and $g = [g, g']j_1 = [g, g']j_2 = g'$. Parallel zigzags are then equal too, so the localization is a preorder, and by Theorem 5.1 it is $\preceq$. With products use the projections and the diagonal. $\square$

Corollary 5.3. *In a topos, and in every presheaf category, $\mathcal{E}[W_{\text{cyc}}^{-1}]$ is the hom-preorder. For directed graphs it is the homomorphism preorder, whose partial-order reflection is the homomorphism order of [5]; two finite graphs are identified iff their cores are isomorphic. An EI-category with binary (co)products is a preorder; a groupoid with binary coproducts is equivalent to a discrete category.*

Proposition 5.4 (natural samenesses). *Call $C \mapsto W(C)$ natural if $F(W(C)) \subseteq W(D)$ for all functors F . Then $\{\text{id}\}$, Iso , W_{obj} (source isomorphic to target), W_{cyc} , Mor , the endomorphism class W_{end} and its intersection with Iso are natural; Iso and W_{end} are incomparable with join W_{obj} ; and every natural sameness other than Mor is contained in W_{cyc} .*

Proof. Each listed class is preserved by every functor (functors preserve identities, isomorphisms, endomorphisms and the existence of a morphism back), and each is a sameness: for W_{obj} and W_{cyc} this is the object-determined criterion (a class of the form $\{f: X \rightarrow Y : X R Y\}$ for an equivalence relation R on objects is a sameness), for W_{end} two-out-of-three holds because a composite of two morphisms is an endomorphism iff both are, when one of them is. An isomorphism between distinct objects lies in Iso but not in W_{end} , and a non-invertible endomorphism in W_{end} but not in Iso , so the two are incomparable; both lie in W_{obj} , and conversely $f: X \rightarrow Y$ with an isomorphism $\varphi: X \rightarrow Y$ is $\varphi \circ (\varphi^{-1}f)$ with $\varphi^{-1}f$ an endomorphism, so $W_{\text{obj}} = \text{Iso} \vee W_{\text{end}}$. Finally let W be natural and $f: X \rightarrow Y$ in $W(C)$ with $Y \not\leq X$; for any morphism $u: A \rightarrow B$ of any category D there is a functor F from the category generated by f (an acyclic category with two objects, since $Y \not\leq X$ and the only endomorphisms are identities) to D with $F(f) = u$, so $u \in W(D)$ and $W = \text{Mor}$. Hence a proper natural sameness contains only morphisms with a back, i.e. $W \subseteq W_{\text{cyc}}$. $\square$

Proposition 5.5. *In a topos, $\text{Loc}(W_{\text{cyc}})$ is the class of subterminal objects and $\text{Sat}(W_{\text{cyc}})$ the class of maps between objects with the same support; W_{cyc} is saturated in **Set** and in **sSet** but not in graphs.*

Proof. If U is subterminal then $\text{Hom}(X, U)$ has at most one element and is non-empty iff $X \rightarrow U$ exists; for $f: X \rightarrow Y$ with a back, $X \rightarrow U$ exists iff $Y \rightarrow U$ exists, so $\text{Hom}(f, U)$ is bijective and U is local. Conversely let Z be local and apply $\text{Hom}(-, Z)$ to the projection $p_1: Z \times Z \rightarrow Z$, which has the back Δ : bijectivity means every map $Z \times Z \rightarrow Z$ factors through p_1 ;

for $p_2 = hp_1$ composing with Δ gives $h = \text{id}$, so $p_1 = p_2$ and $Z \rightarrow 1$ is a monomorphism. Thus $\text{Loc}(W_{\text{cyc}}) = \text{Sub}(1)$. A map $f: X \rightarrow Y$ is orthogonal to all subterminals iff for every subterminal U a map $X \rightarrow U$ exists iff a map $Y \rightarrow U$ exists, i.e. iff X and Y have the same support (the support of X being the least U with $X \rightarrow U$). Maps with a back preserve supports, so $W_{\text{cyc}} \subseteq \text{Sat}(W_{\text{cyc}})$, with equality iff any two objects with the same support admit morphisms in both directions: true in **Set** and in **sSet** (all non-empty objects have support 1 and admit maps both ways), false for graphs (the graph with one vertex and no edge and the graph with one vertex and a loop have the same support, but there is no morphism from the looped graph to the loopless one). $\square$

6. LIMITS, COUNTEREXAMPLES AND OPEN PROBLEMS

6.1. Joins are not preserved in general.

Theorem 6.1. *On the poset $\mathbb{N}$ let $J_{\text{ev}}, J_{\text{od}}$ be the topologies of the ideals of morphisms passing through an even, resp. odd, number. Their join is the degenerate topology, so $W_{J_{\text{ev}} \vee J_{\text{od}}} = \text{Mor}$, but $W_{\text{ev}} \vee W_{\text{od}}$ is contained in the class of maps inverted by $\lim_{\mathbb{N}^{\text{op}}}$, which does not contain the map from the presheaf of final segments to the terminal presheaf.*

Proof. Degeneracy: $\emptyset$ covers 0 for J_{od} , and inductively the least covering sieve of m for the topology of the other parity consists of arrows from smaller objects, which are covered by $\emptyset$; transitivity gives $\emptyset \in J(m)$. $W_{\text{ev}} = \{f : f_y \text{ bijective for even } y\}$ and the evens are cofinal, so $\lim_{\mathbb{N}^{\text{op}}}$ inverts W_{ev} ; likewise W_{od} ; hence it inverts their join. For $X(n) = \{x \geq n\}$ the limit is empty, while the limit of the terminal presheaf is a point. $\square$

6.2. A refuted conjecture. [16, Conj. 6.12] proposed that for idempotent ideals $X, Y, Z \neq M$ of a finite monoid the multiplication $X \otimes Y \otimes Z \rightarrow XY \otimes Z$ is always bijective, which by Lemma 3.4 would give join preservation for all finite monoids. It holds for all monoids of order ≤ 5 but fails in general.

Proposition 6.2. *Let P be the diamond poset $0 < 1, 2 < 3$ (with 1, 2 incomparable) and M its contracted monoid: an identity, the nine morphisms $a \rightarrow b$ ($a \leq b$), and a zero, with product the composite when defined and 0 otherwise. Let $X = Y$ be the ideal of all morphisms except the identities of 0 and 3, and $Z = M 1_3 M$, all proper idempotent ideals. Then $[0 \rightarrow 1, 1 \rightarrow 3] \neq [0 \rightarrow 2, 2 \rightarrow 3]$ in $X \otimes_M Y$ although both have product $0 \rightarrow 3$, and the classes $[0 \rightarrow 1, 1 \rightarrow 3, 1_3]$ and $[0 \rightarrow 2, 2 \rightarrow 3, 1_3]$ remain distinct in $X \otimes Y \otimes Z$, so $X \otimes Y \otimes Z \rightarrow XY \otimes Z$ is not injective.*

Proof. Write $x = 0 \rightarrow 1$, $y = 1 \rightarrow 3$. A tensor move replaces (x, y) by (x', my) with $x = x'm$, $x' \in X$, or by (xm, y') with $y = my'$, $y' \in Y$. The factorizations of x in M are $x \cdot 1_1$ and $1_0 \cdot x$ with $1_0 \notin X$, and those of y are $1_1 \cdot y$ and $y \cdot 1_3$ with $1_3 \notin Y$; so all moves are trivial and the class of (x, y) is the singleton $\{(x, y)\}$; likewise for $(0 \rightarrow 2, 2 \rightarrow 3)$. Let $Z = M 1_3 M =$

$\{0 \rightarrow 3, 1 \rightarrow 3, 2 \rightarrow 3, 1_3, 0\}$, an idempotent proper ideal, and $c = 1_3$. In $X \otimes Y \otimes Z$ the additional moves act on the last two slots: $(y, c) \mapsto (y', mc)$ with $y = y'm$, $y' \in Y$, and $(y, c) \mapsto (ym, c')$ with $c = mc'$, $c' \in Z$. Since $1_1 \in Y$, the factorization $y = 1_1 \cdot y$ gives the move $(x, y, 1_3) \mapsto (x, 1_1, y)$, and from $(x, 1_1, y)$ the only non-trivial move is the inverse one; so the class of (x, y, c) is $\{(x, y, 1_3), (x, 1_1, y)\}$, and likewise the class of $(0 \rightarrow 2, 2 \rightarrow 3, 1_3)$ is $\{(0 \rightarrow 2, 2 \rightarrow 3, 1_3), (0 \rightarrow 2, 1_2, 2 \rightarrow 3)\}$. The two classes are disjoint but both map to $[0 \rightarrow 3, 1_3]$ in $XY \otimes Z$. $\square$

Remark 6.3. The conjecture was a sufficient condition applied to all triples of ideals, whereas only the triples along the alternating pattern $D, E, D, E, \dots$ enter the tower; a finite enumeration suggests that on this monoid the towers of all pairs stabilize, so the conjecture is stronger than what join preservation requires.

6.3. Essential constancy for finite categories. The obstruction in Theorem 6.1 is a pro-limit invariant of the alternating plus-construction tower. For finite categories no such invariant exists. We use the Yoneda lemma for pro-objects: for a small category $\mathcal{A}$, the functor $\text{Pro}(\mathcal{A})^{\text{op}} \rightarrow \text{Fun}(\mathcal{A}, \mathbf{Set})$, $(T_k)_k \mapsto \text{colim}_k \text{Hom}(T_k, -)$, is fully faithful, so that $\text{Hom}_{\text{Pro}(\mathcal{A})}((T_k)_k, X) = \text{colim}_k \text{Hom}(T_k, X)$ for $X \in \mathcal{A}$ and two pro-objects with naturally isomorphic functors are isomorphic [8, §6.1].

Theorem 6.4. *Let M be a finite monoid, D, E idempotent ideals, I' the largest idempotent ideal in $D \cap E$, and $m \geq 1$ such that $(DE)^m = (ED)^m = I'$ (the descending chains $(DE)^j, (ED)^j$ stabilize at idempotent ideals contained in $D \cap E$, which contain $I' = I'^{2j}$, hence equal I'). Put $P = I' \otimes_M I'$, so that $a_{I'}X = \text{Hom}(P, X)$. For every M -set X , $\text{colim}_k \Phi_k X \cong a_{I'}X$ naturally; hence the pro-object (T_k, μ_k) is pro-isomorphic to the constant object P , and for k large $a_{I'}X$ is a natural retract of $\Phi_k X$ compatibly with the units. Moreover $\text{im}(T_{k+l} \rightarrow T_k) = I' \cdot T_k$ for $l \geq 2m$, so the tower stabilizes (μ_k bijective for large k) iff $T_k = I' \cdot T_k$ for large k , iff every class of T_k has a representative whose first slot lies in I' ; in that case $W_D \vee W_E = W_{I'}$.*

Proof. Let $L = \text{colim}_k \Phi_k X$. Since D is a finite M -set, $\text{Hom}_M(D, -)$ commutes with filtered colimits, so $L^{+D} = \text{colim}_k (\Phi_k X)^{+D} = \text{colim}_{D_{k+1}=D} \Phi_{k+1} X = L$ with the unit of L the identity of the colimit; thus L is a D -sheaf, likewise an E -sheaf, hence an I' -sheaf. The maps $X \rightarrow \Phi_k X$ are composites of maps of $W_D \cup W_E$, hence lie in $W_D \vee W_E \subseteq W_{I'}$, and $a_{I'} = \text{Hom}(P, -)$ commutes with filtered colimits, so $a_{I'}L = a_{I'}X$ and $L = a_{I'}L = a_{I'}X$; no finiteness of X is used. Restricting to finite X , the functors $\text{colim}_k \text{Hom}(T_k, -)$ and $\text{Hom}(P, -)$ on finite M -sets are naturally isomorphic, and by the Yoneda lemma for pro-objects $(T_k)_k \cong P$ in $\text{Pro}(\text{Fin}M)$. Explicitly, the isomorphism consists of compatible maps $b_k: P \rightarrow T_k$ and a map $a: T_{k_0} \rightarrow P$ with $ab_{k_0} = \text{id}_P$ and, for each j and all $k \gg j$, $b_j a \mu_{k_0}^k = \mu_j^k: T_k \rightarrow T_j$; applying $\text{Hom}(-, X)$ gives the retraction $\rho_k = b_k^*$ of $\sigma_k = (a \mu_{k_0}^k)^*: a_{I'}X \rightarrow \Phi_k X$

and the factorization of $\Phi_{k_0}X \rightarrow \Phi_kX$ through $a_{I'}X$. Finally the image of $T_{k+l} = D_{k+l} \otimes \cdots \otimes D_{k+1} \otimes T_k$ in T_k is $(D_{k+l} \cdots D_{k+1}) \cdot T_k = I' \cdot T_k$ for $l \geq 2m$, since the product of $2m$ alternating ideals is I' ; if $T_k = I' T_k \subseteq D_{k+1} T_k$ then μ_{k+1} is surjective, the sizes $|T_k|$ are eventually non-increasing, hence constant, and μ_k is eventually bijective; conversely eventual bijectivity gives $T_k = \text{im}(T_{k+2m} \rightarrow T_k) = I' T_k$. In that case Lemma 3.2 gives $W_D \vee W_E = W_{I'}$. $\square$

Corollary 6.5 (finite categories). *Theorem 6.4 holds for a finite category S in place of M , with D, E idempotent two-sided ideals of S , $\otimes_M$ replaced by the tensor product of S - S -bimodules (functors $S^{\text{op}} \times S \rightarrow \mathbf{Set}$), and $\text{Hom}_M(T, X)$ replaced by the presheaf $c \mapsto \text{Hom}_{\text{Presh}(S)}(T(-, c), X)$.*

Proof. The ideals D, E, I' and the tensors $T_k, P = I' \otimes_S I'$ are finite bimodules (finitely many morphisms in each value), $X^{+D}(c) = \text{Hom}(D(-, c), X)$ is the plus construction for the covering sieve $D(-, c)$, and $a_{I'}X = \text{Hom}(P, X)$ by [16, Thm. 6.10]. Each $D(-, c)$ and $T_k(-, c)$ is a finite presheaf, so $\text{Hom}(D(-, c), -)$ and $\text{Hom}(T_k(-, c), -)$ commute with filtered colimits, and the first paragraph of the proof applies verbatim, objectwise in c and for every presheaf X : $L = \text{colim}_k \Phi_k X$ is a D -sheaf and an E -sheaf, and $L \cong a_{I'}X$. The functor $T \mapsto \text{Hom}(T, -)$ from finite bimodules to functors $\text{FinPresh}(S) \rightarrow \text{Presh}(S)$ is fully faithful: for each c the functor $X \mapsto \text{Hom}(T(-, c), X)$ on finite presheaves is represented by the finite presheaf $T(-, c)$, so the Yoneda lemma on $\text{FinPresh}(S)$ recovers $T(-, c)$ together with the maps between such functors, and naturality in c recovers the bimodule structure. Hence the Yoneda lemma for pro-objects applies in $\text{Pro}(\text{finite bimodules})$ and gives $(T_k)_k \cong P$ with the same retract structure. Finally the product of $l \geq 2m$ consecutive alternating ideals is I' , so $\text{im}(T_{k+l} \rightarrow T_k) = I' \cdot T_k$ as before, and Lemma 3.2 holds for presheaves on S with the same proof. $\square$

6.4. The tower need not stabilize.

Proposition 6.6. *Let $\mathcal{C}$ be the category with two objects a, b , arrows $u: a \rightarrow b, v: b \rightarrow a$, freely generated subject to the identification of all composites of length ≥ 3 with the same endpoints; it has 10 arrows. Let M be its contracted monoid (12 elements), D the ideal of arrows factoring through b and E that through a . Then $T_k \neq I' \cdot T_k$ for every even k : the alternating plus-construction tower of (D, E) never stabilizes.*

Proof. The identification of long words is a congruence because lengths add under composition, so $\mathcal{C}$ is a category with, for each ordered pair of objects, the words of length ≤ 2 between them and one long arrow; D and E are idempotent (they contain identities), proper and incomparable, and I' is the ideal of long arrows and 0. By Theorem 6.4 it suffices to exhibit, for every even k , a class of T_k with no representative whose first slot lies in I' . Take $k = 2j$ and $t = (1_a, uv, 1_a, uv, \dots, 1_a, uv) \in T_k$, whose slots alternate

between E and D as required. Every tuple of short arrows with product the word $w = (uv)^j$ of the free category is a *cut configuration* of w : cut points $0 = c_0 \leq c_1 \leq \dots \leq c_k = 2j$, the i -th slot being the subword between c_{i-1} and c_i , and an empty slot being the identity of the object at its cut point, which must be a (even cut point) for an E -slot (i odd) and b (odd cut point) for a D -slot (i even). A tensor move between two tuples of short arrows moves one cut point and preserves w . Put $d_i := i - c_i$, so $d_0 = d_k = 0$, a slot of length ℓ changes d by $1 - \ell$, and an empty slot raises d by one and is allowed only when d_i is odd, i.e. only from an even value of d . From an odd value d cannot rise, since the next slot must be non-empty; hence $d \leq 1$ throughout, and once $d \leq -1$ it can never return to 0 (from an even value it rises by one to an odd value and is then stuck). A slot of length ≥ 3 would lower d from at most 1 to at most -1 , so no cut configuration of w has a slot of length ≥ 3 . Consequently no tensor move applied to a cut configuration produces a long slot (the target would be a cut configuration with a slot of length ≥ 3) or a zero slot (adjacent slots of a configuration are composable), and the class of t consists of cut configurations only; its first slot is never in I' . $\square$

Remark 6.7. Enumerating the tensor relations gives $|T_1| = 10$ and $|T_k| = 9$ for $k \geq 2$, with $|I'T_k| = 5$: four classes of short tuples survive at every level. For the analogous monoid with three objects in a cycle and words of length ≥ 4 identified ($|M| = 23$) the sizes oscillate, 19, 19, 16, 19, 16, $\dots$

Whether $W_D \vee W_E = W_{J_D \vee J_E}$ holds on the monoids of Proposition 6.6 is not decided by Lemma 3.2, which is only sufficient; by Theorem 6.4 no pro-limit of the tower can obstruct it, and the dual tower $\lim_k X \otimes_M (D \otimes D) \otimes (E \otimes E) \otimes \dots$, which inverts W_D and W_E , is $X \otimes P$ and does not obstruct it either.

Question 6.8. Is $\text{Top}(S)$ a sublattice of $\text{Same}(\text{Presh}(S))$ for every finite category S ? Equivalently, is the unit $X \rightarrow a_{J \vee J'} X$ in $W_J \vee W_{J'}$ for all X ? The first undecided cases are the monoids of Proposition 6.6.

Question 6.9. Classify the natural samenesses (Proposition 5.4); is the interval $[\text{Iso}, W_{\text{cyc}}]$ of natural samenesses finite?

Question 6.10. Does Theorem 1.1 hold for cubical objects of an ∞ -topos with the cubical coskeleta? For cubical sets it does [15]; the missing ingredient is a cubical form of Proposition 4.3.

FUNDING

No funding was received for this work.

DECLARATIONS

CRedit authorship contribution statement. Ryota Shibusawa: Conceptualization, Methodology, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Competing interests. The author declares that he has no competing interests.

REFERENCES

- [1] M. Anel, G. Biedermann, E. Finster, A. Joyal, *Left-exact localizations of ∞ -topoi I: Higher sheaves*, Adv. Math. **400** (2022), 108268.
- [2] M. Anel, G. Biedermann, E. Finster, A. Joyal, *Left-exact localizations of ∞ -topoi II: Grothendieck topologies*, J. Pure Appl. Algebra **228** (2024), 107597.
- [3] J. Duskin, *Simplicial methods and the interpretation of “triple” cohomology*, Mem. Amer. Math. Soc. **163** (1975).
- [4] P. G. Goerss, J. F. Jardine, *Simplicial Homotopy Theory*, Progress in Mathematics **174**, Birkhäuser, 1999.
- [5] P. Hell, J. Nešetřil, *Graphs and Homomorphisms*, Oxford Lecture Series in Mathematics and its Applications **28**, Oxford University Press, 2004.
- [6] P. T. Johnstone, *Sketches of an Elephant: A Topos Theory Compendium*, Oxford University Press, 2002.
- [7] W. Lück, *Transformation Groups and Algebraic K-Theory*, Lecture Notes in Mathematics **1408**, Springer, 1989.
- [8] M. Kashiwara, P. Schapira, *Categories and Sheaves*, Grundlehren der mathematischen Wissenschaften **332**, Springer, 2006.
- [9] G. M. Kelly, F. W. Lawvere, *On the complete lattice of essential localizations*, Bull. Soc. Math. Belg. Sér. A **41** (1989), 289–319.
- [10] C. Kennett, E. Riehl, M. Roy, M. Zaks, *Levels in the toposes of simplicial sets and cubical sets*, J. Pure Appl. Algebra **215** (2011), 949–961.
- [11] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies **170**, Princeton University Press, 2009.
- [12] J. Marquès, *A criterion for categories on which every Grothendieck topology is rigid*, Appl. Categ. Structures **33** (2025), art. 37.
- [13] S. Mac Lane, I. Moerdijk, *Sheaves in Geometry and Logic*, Springer, 1992.
- [14] R. Shibusawa, *The sameness lattice of a category: weak-equivalence structures, Galois functoriality, and comparison intervals*, preprint (2026), archived at DOI 10.5281/zenodo.22823772.
- [15] R. Shibusawa, *Postnikov truncations as joins of sheaf and homotopy equivalences*, preprint (2026), archived at DOI 10.5281/zenodo.22956181.
- [16] R. Shibusawa, *Meets and joins of Grothendieck topologies in the sameness lattice*, preprint (2026), archived at DOI 10.5281/zenodo.22978446.
- [17] R. Shibusawa, *The EI-reflection of a category, its preorder reflection, and the homomorphism order of graphs*, preprint (2026), archived at DOI 10.5281/zenodo.23029859.
- [18] R. Shibusawa, *Postnikov truncation of simplicial objects in an ∞ -topos as a join of localizations*, preprint (2026), archived at DOI 10.5281/zenodo.23049588.

DAIICHI INSTITUTE OF TECHNOLOGY

Email address: `r-shibusawa@daichi-koudai.ac.jp`